Ham Radio Weather Dashboard

A weather dashboard built for amateur radio operators

Version 1.0.10 · Qt6 / PySide6 · Free & open source · github.com/N8SDR1/ham-radio-weather

What it is

A modern desktop weather application built specifically for amateur radio operators. Pulls live data from your Ambient or Ecowitt station and layers ham-specific overlays on top — HF propagation, amateur-satellite passes, lightning proximity warnings, and NOAA geomagnetic-storm indicators — in a single drag-to-reorder dashboard. A "None" mode runs the entire app on free online feeds for operators without a station.

Weather tiles

- **Outdoor**, **Wind** (360° compass), **Rain**, **Lightning**, **Indoor**
- **Humidity**, **UV**, **Solar Radiation**, **Pressure**
- **24 h sparklines** + **3 h pressure trend arrow** on Ambient *and* Ecowitt
- **Dramatic mood effects** *(new in 1.0.10)* — fire/ice halos, zigzag lightning bolts, rate-scaled rain
- **7-day Forecast** + **Sun / Moon** (locally computed)
- Air Quality, Soil, Leak — auto-populate on sensor detect

Ham-radio overlays

- **HF Propagation** — SFI, K/A, band-condition grid, G-scale badge
- **Space Weather** — 72 h Kp forecast + aurora ribbons at G4/G5
- **Satellites** — 13 birds, SGP4 pass prediction via Celestrak TLEs
- "Unplug the Rig!" lightning panic, "Antenna Swayer!" needle wobble
- **Tile Personality** *(new in 1.0.10)* — configurable mood thresholds

Dashboard experience

- Dark / light / auto themes · °F / °C toggle · drag-to-reorder with per-tile S/M/L/XL sizing
- Configurable alert thresholds (heat, freeze, high wind, lightning nearby, fire weather, shack overheat)
- Header update-pill pulses automatically when a new GitHub release is published

Weather station support

Brand / mode	Connection	Status
Ambient Weather (WS-2000/2902/5000/1965)	REST polling + Socket.IO realtime	**Fully supported**
Ecowitt (Wittboy, GW/HP gateways)	Ecowitt Cloud API v3 (60 s poll)	**Fully supported**
None — no station required	Open-Meteo + Blitzortung + NOAA SWPC + HamQSL	**Fully supported**
Tempest / Davis / Netatmo / Ecowitt LAN	Various (see roadmap)	Planned

Running without a weather station

Pick **None** in Settings → Weather Station and enter a Maidenhead grid square. Open-Meteo provides current conditions and the 7-day forecast; Blitzortung provides real-time lightning filtered to a user-configurable radius; HamQSL and NOAA SWPC cover HF propagation and space weather; Celestrak TLEs drive satellite passes. Station-only tiles (Indoor shack, battery) auto-hide so nothing shows empty.

Active severe-weather alerts

Two layers side by side: user-configurable **threshold rules** evaluated against live data, plus **official regional feeds** selected in Settings → Alerts:

Region	Provider	Feed
United States	National Weather Service (NWS)	api.weather.gov CAP alerts
Canada	Environment and Climate Change Canada	Public CAP feed per province
Europe	MeteoAlarm	Pan-European severe-weather aggregator
Australia	Bureau of Meteorology	Public warnings feed per state

How to obtain it

Windows: download the latest *HamRadioWeather-Setup-X.Y.Z.exe* installer from **github.com/N8SDR1/ham-radio-weather/releases**. Per-user install, no admin required. Accept the disclaimer on first run, pick your station (or None mode), and go. **macOS / Linux:** runs from source with Python 3.11+ — see the project README. **License:** free and open source; optional PayPal donation linked from the About dialog.

Built by N8SDR — Rick Langford, Hamilton OH (grid EM79RJ). Free to use, free to share. 73.